DD-F-786c

NOVEMBER 4, 1960

SUPERSEDING
Int. Fed. Spec. DD–F–00786b (HEW–PHS)
August 14, 1959 and
Fed. Spec. DD–F–786a
October 10, 1952

FEDERAL SPECIFICATION

FUNNEL, FILTERING, LABORATORY, PORCELAIN, BUCHNER TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers one grade of Buchner type porcelain filtering funnels.

1.2 Classification.

1.2.1 *Sizes.*—The filtering funnels covered by this specification shall be of the following sizes, as specified (see 6.1):

Size	*Outside diameter*
	Millimeter
0	51
1	67
2A	106
4	142
4A	171
5	213

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–B–676 — Boxes, Set-Up, Paperboard.
PPP–C–843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. — The material shall be of such quality porcelain that it shall pass the tests specified herein.

3.2 Shock resistance.—No cracking shall occur when the funnels are subjected to the shock test described in 4.4.1.

3.3 Glaze.—The funnels shall be glazed, inside and outside, except for the rim.

3.3.1 *Resistance to chemical attack.*—Resistance to chemical attack as measured by loss in weight by boiling in a solution of sodium hydroxide shall be not more than 0.144 percent when tested as described in 4.4.2.

3.4 Style and design.—The funnels shall be the conventional Buchner type, having a permanently fixed perforated filter plate at the bottom of the maximum diameter of the straight sides, which shall straight taper below the plate, directly to the stem. The end of the stem shall be beveled.

3.5 Dimensions.—The dimensions of the funnels shall conform to table I.

3.6 Identification marking.—The funnels shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.7 Workmanship.—The funnels shall be symmetrical in shape, uniform in size, and free from uneven porcelain distribution, cracks, specks, and other irregularities in the body or glaze which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *For visual examination.* — Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2.2 *For tests.*—Sample size for each test shall be as indicated for inspection level L4 in Military Standard MIL–STD–105, except that a minimum of five (5) samples shall be taken.

4.3 Inspection.—The samples selected in accordance with 4.2.1 shall be inspected for construction, style, design, dimensions, marking, and workmanship requirements, as specified in section 3. The Acceptable Quality Level (AQL) shall be 4.0 percent defective. The unit of product for visual examination shall be one funnel. Any departure from the construction, style, design, marking, and workmanship requirements shall be considered a defect. Any departure from the dimensional

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TABLE I.—*Dimensions*

Size	Outside diameter	Inside diameter at perforated plate	Diameter perforated area	Number of perforations	Distance rim to perforated plate	Height overall	Length of tube	Diameter of tube
	Millimeters	*Millimeters*	*Millimeters*		*Millimeters*	*Millimeters*	*Millimeters*	*Millimeters*
0	51±3	42+5 −1	32±2	30 to 40	24±2	89±3	46±3	13±2
1	67±3	56+5 −1	45±3	75 to 125	30±2	101±3	47±3	14±2
2A	106±4	91+7 −1	80±3	100 to 150	46±3	160±10	71±10	20±2
4	142±5	126+8 −1	115±5	110 to 160	59±3	202±10	88±10	22±2
4A	171±5	151+9 −1	139±5	200 to 250	61±4	221±10	103±10	25±2
5	213±5	186+9 −1	173±5	250 to 300	70±4	270±10	115±10	26±2

requirements not within the specified tolerances shall be considered a defect.

4.4 Tests.—The samples selected in accordance with 4.2.2 shall be tested as specified in 4.4.1 and 4.4.2. Any departure from the test requirements shall be considered a defect. The acceptance number shall be zero (0) for all sample sizes.

4.4.1 *Shock test.* — Test samples shall be heated to 160° ± 2°C., in an oven for 30 minutes, removed, and immediately plunged into water having a temperature of 10° ± 2°C.

4.4.2 *Resistance to chemical attack.*—The funnels shall be completely immersed in 10 percent sodium hydroxide and kept at boiling temperature for 8 hours. After removal from the sodium hydroxide solution, the funnels shall be washed in water and diluted hydrochloric acid, then rinsed and dried in an oven and weighed.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Each funnel shall be packaged in a carton of appropriate size conforming to the requirements of Federal Specification PPP–B–636, type I or II, class 2, or Federal Specification PPP–B–676, type I, II, or III, class A, B, or C, style 1 or 2. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material conforming to Federal Specification PPP–C–843 to insure against shifting of, or damage to, the contents. Carton shall be adequately secured to prevent accidental opening.

5.1.1.1 *Intermediate packaging.* — The quantity of funnels specified in table II, pack-

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aged as specified in 5.1.1, shall be placed in an intermediate container conforming to Federal Specification PPP–B–636, type I, class 1. Carton shall be adequately secured to prevent accidental opening.

5.1.2 *Level C.*—Unless otherwise specified, funnels shall be packaged in accordance with accepted commercial practice.

5.2 Packing.

5.2.1 *Level A.*—The quantity of funnels specified in table II, packaged as specified in 5.1.1.1, shall be overpacked in a wood box conforming to Federal Specification PPP–B–621, class 2. Unless otherwise specified, each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.* — The quantity of funnels specified in table II, packaged as specified in 5.1.1.1, shall be overpacked in a wood box conforming to Federal Specification PPP–B–621, class 1.

5.2.3 *Level C.*—The funnels shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.2.4 *Packing variation permitted, applicable to level A or level B.*—If the required number of funnels to be shipped is less than the number specified to be overpacked in an exterior container, such funnels shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required by the contract or order, marking shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Interior packages.*—Each individual package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

TABLE II.—*Packaging quantities*

Size	Quantity of funnels to be packaged in each intermediate carton	Quantity of intermediate cartons to be overpacked in each shipping container	Total number of funnels in each shipping container
0	12	6	72
1	8	6	48
2A	4	6	24
4	Not required	Not required	12
4A	Not required	Not required	12
5	Not required	Not required	12

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6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size required (see 1.2.1).
(*c*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Transportation description.—Transportation description applicable to this item is:

Porcelainware, not otherwise indexed by name. (Released valuation exceeding $4.50 per pound *not taken.* Ship via parcel post or express.)

Carload minimum weights 20,000 and 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Porcelainware, not otherwise indexed. (Released valuation not to exceed $20.00 per 100 pounds.)

Motor volume minimum weight 24,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O
Navy—Md.

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